

Feature-Selection-Based Koopman Modeling and Robust LQR Control of Robots

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ABSTRACT

Extended Dynamic Mode Decomposition (EDMD) provides a practical route to Koopman-based modeling by representing nonlinear robot dynamics as linear evolution in a lifted space. Its accuracy, however, depends strongly on the chosen dictionary: a small dictionary may miss important nonlinearities, while an overcomplete one can introduce redundancy, ill-conditioning, and unnecessary computation. We propose a feature-selection-based Koopman representation learning framework that retains the interpretability of a physics-informed candidate library while learning a compact set of linear feature combinations. The selection matrix is optimized jointly with the Koopman matrices, and the resulting problem is solved by block coordinate descent with stationary-point convergence guarantees under standard assumptions. The learned lifted model is then used in a feedforward-feedback linear quadratic regulator that treats finite-dimensional approximation errors and external uncertainties as bounded lumped disturbances. Lyapunov analysis shows that the closed-loop tracking error is ultimately uniformly bounded and relates the disturbance level to the steady-state error bound. The method is evaluated on a three-degree-of-freedom rigid robot arm and a cable-driven soft platform. The proposed feature-selection-based EDMD (FS-EDMD) model achieves lower multi-step prediction errors than EDMD and Extended Dynamic Mode Decomposition with Dictionary Learning (EDMDDL), and the resulting controller reduces tracking error relative to a PID baseline over multiple references. A disturbance-recovery test on the robotic arm further demonstrates bounded deviations and smooth recovery.

1. Introduction

Modern robotic systems, including manipulators, mobile robots, and soft robots, exhibit strongly nonlinear dynamics due to mechanical coupling, actuator effects, and uncertain interactions with the environment [5, 18]. Accurate dynamic models are essential for high-performance control, yet first-principles modeling becomes difficult when a robot contains flexible structures, uncertain parameters, or unmodeled interactions [14]. Data-driven models learned from operational data therefore provide an attractive alternative for robotic prediction and control [3, 21, 12, 11].

Koopman operator theory offers a control-oriented way to represent nonlinear dynamics by linear evolution in a lifted observable space [7, 9]. Since the Koopman operator is infinite-dimensional, practical methods rely on finite-dimensional approximations. EDMD constructs such an approximation by lifting the state through a user-specified dictionary of basis functions [24, 9]. The resulting lifted linear models enable the use of linear-system tools, including Koopman-based model predictive control and Koopman-based linear quadratic regulation, for nonlinear robotic systems [16, 8, 4].

A key bottleneck of EDMD is the choice of basis functions. In principle, the dictionary should span a subspace that is approximately invariant under the Koopman operator. In practice, constructing such a dictionary requires substantial prior knowledge of the underlying dynamics [15]. This leads to a tradeoff: a small dictionary may miss important

nonlinear features, whereas an overcomplete dictionary increases computational cost, introduces redundancy and ill-conditioning, and may cause overfitting [14, 20].

Two common strategies are learned dictionaries and sparse selection. Deep learning methods can learn nonlinear coordinates from data [13, 19, 17], but the resulting features are often difficult to interpret and analyze in robotic control. For example, EDMDDL learns a nonlinear dictionary for the Koopman operator [10]. Sparse regression methods select compact representations from large libraries [25], but they may involve nonsmooth optimization and sensitivity to noise and hyperparameters [23]. For robotic systems, it is therefore desirable to retain the interpretability of physics-informed candidate functions while avoiding the computational burden of using the full overcomplete dictionary.

We therefore retain an overcomplete, physics-informed candidate library and learn a lower-dimensional recombination of its functions. An optimizable selection matrix parameterizes this recombination, and the resulting FS-EDMD model is trained directly for lifted one-step prediction. Unlike PCA, which preserves dominant data variance, our objective targets the prediction error that matters for control. The Koopman and selection matrices are then updated jointly by block coordinate descent with convergence guarantees. Using the learned lifted model, we design a feedforward-feedback Koopman-LQR controller for trajectory tracking. The finite-dimensional approximation error is treated as a bounded disturbance, allowing the closed-loop tracking error to be analyzed through Lyapunov arguments.

The main contributions are summarized as follows.

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- (1) We propose a feature-selection-based framework for Koopman representation learning in robotic systems. A physics-informed overcomplete candidate library is constructed, and a learnable feature selection matrix is embedded into the lifted state. This yields compact and informative feature combinations while preserving the interpretability of the candidate library.
- (2) The Koopman modeling task is formulated as a joint optimization problem over the Koopman matrices and the feature selection matrix. The resulting problem is solved by a block coordinate descent algorithm, where the matrix update subproblems are explicitly characterized and convergence to a stationary point is established under standard assumptions.
- (3) A robust feedforward-feedback Koopman-LQR controller is developed for robotic trajectory tracking. By modeling finite-dimensional Koopman approximation errors and external uncertainties as bounded lumped disturbances, Lyapunov analysis proves that the closed-loop tracking error is ultimately uniformly bounded and relates disturbance level to the steady-state tracking bound.
- (4) The proposed modeling and control framework is validated on two robotic platforms with distinct dynamics: a three-degree-of-freedom rigid robot arm and a cable-driven soft platform. The experiments evaluate multi-step model prediction, trajectory tracking on several reference patterns, and disturbance recovery.

The rest of this paper is organized as follows. Section 2 introduces the preliminaries of Koopman operator theory and formulates the problem considered in this paper. Section 3 presents the feature-selection-based Koopman representation learning framework and the corresponding block coordinate descent algorithm. Section 4 develops the Koopman-based feedforward-feedback LQR controller and analyzes the closed-loop robust tracking performance. Section 5 reports prediction and control experiments on both robotic platforms. Finally, Section 6 concludes this paper.

Notation: Unless otherwise stated, $\|\cdot\|$ denotes the Euclidean norm for vectors and the induced 2-norm for matrices, $\|\cdot\|_1$ denotes the ℓ_1 norm, and $\|\cdot\|_F$ denotes the Frobenius norm for matrices; $\mathbb{R}^n$ denotes the n -dimensional Euclidean space; the superscripts $(\cdot)^\top$ and $(\cdot)^\dagger$ denote the transpose and the Moore–Penrose pseudoinverse, respectively; $\mathbf{I}_n$ denotes the $n \times n$ identity matrix; $\mathbf{0}_{m \times n}$ denotes the $m \times n$ zero matrix; $\text{tr}(\cdot)$, $\text{rank}(\cdot)$, and $\text{diag}(\cdot)$ denote the trace, rank, and diagonal matrix operator, respectively; $\mathcal{K}$ denotes the Koopman operator; $\mathcal{H}$ denotes a Hilbert space; $\Phi(\mathbf{x})$ denotes the feature library; for a matrix $\mathbf{A}$, $\mathbf{A} \geq 0$ and $\mathbf{A} > 0$ mean that $\mathbf{A}$ is positive semidefinite and positive definite, respectively.

2. Preliminaries and problem formulation

2.1. Koopman Operator Theory

Consider the discrete-time nonlinear system

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k), \quad (1)$$

where $\mathbf{x}_k \in \mathbb{R}^n$ is the system state and $\mathbf{u}_k \in \mathbb{R}^m$ is the control input at time step k . The mapping $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ describes the state transition. It is assumed that f is Lipschitz continuous on its domain.

Let $\mathcal{H}$ be a Hilbert space of real-valued observation functions defined on the state space. For any observation function $g \in \mathcal{H}$, when the control input is $\mathbf{u}_k$, the discrete-time Koopman operator $\mathcal{K} : \mathcal{H} \rightarrow \mathcal{H}$ is defined by the evolution of g along the system trajectories as

$$(\mathcal{K}g)(\mathbf{x}_k, \mathbf{u}_k) \triangleq g(f(\mathbf{x}_k, \mathbf{u}_k)) = g(\mathbf{x}_{k+1}). \quad (2)$$

The core of Koopman operator theory is to lift the nonlinear dynamics from the finite-dimensional state space to an infinite-dimensional function space $\mathcal{H}$, thereby achieving linear evolution that obeys the superposition principle:

$$\mathcal{K}(\alpha g_1 + \beta g_2) = \alpha \mathcal{K}g_1 + \beta \mathcal{K}g_2, \quad (3)$$

where $g_1, g_2 \in \mathcal{H}$ are arbitrary observation functions and $\alpha, \beta \in \mathbb{R}$ are arbitrary scalars. This linearity enables the application of linear-system tools to analyze nonlinear dynamics after lifting.

2.2. Problem Formulation

As $\mathcal{H}$ is typically infinite-dimensional, EDMD approximates the Koopman operator in a finite-dimensional space spanned by a candidate function library $\{g_i\}_{i=1}^M$. The approximation quality depends critically on this library. A small library may miss essential nonlinear features, whereas an overcomplete library increases computational cost and may introduce redundancy and overfitting.

This accuracy–complexity tradeoff constitutes a central challenge in basis function selection for EDMD and related data-driven Koopman methods. To address this challenge and enable practical robotic control, this paper aims to: (1) learn a compact Koopman lifted representation from a physics-informed overcomplete candidate function library by using a feature selection matrix; and (2) design a robust feedforward-feedback LQR controller based on the learned lifted model to achieve reliable trajectory tracking under bounded lumped disturbances.

3. Koopman Representation Learning Based on Feature Selection Mechanism

This section develops the feature-selection-based Koopman learning framework, including the joint optimization algorithm and the lumped-disturbance model used for robust control.

3.1. Learnable Feature Selection and Joint Optimization

We first construct an overcomplete candidate observation function library $\Phi(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}^P$, where $P \gg n$, to provide sufficient nonlinear expressiveness. To remove redundant components while preserving the physics-informed structure of the library, a learnable feature selection matrix $\mathbf{D} \in \mathbb{R}^{N \times P}$ with $N \ll P$ is introduced. The resulting lifted coordinate is formally defined as follows.

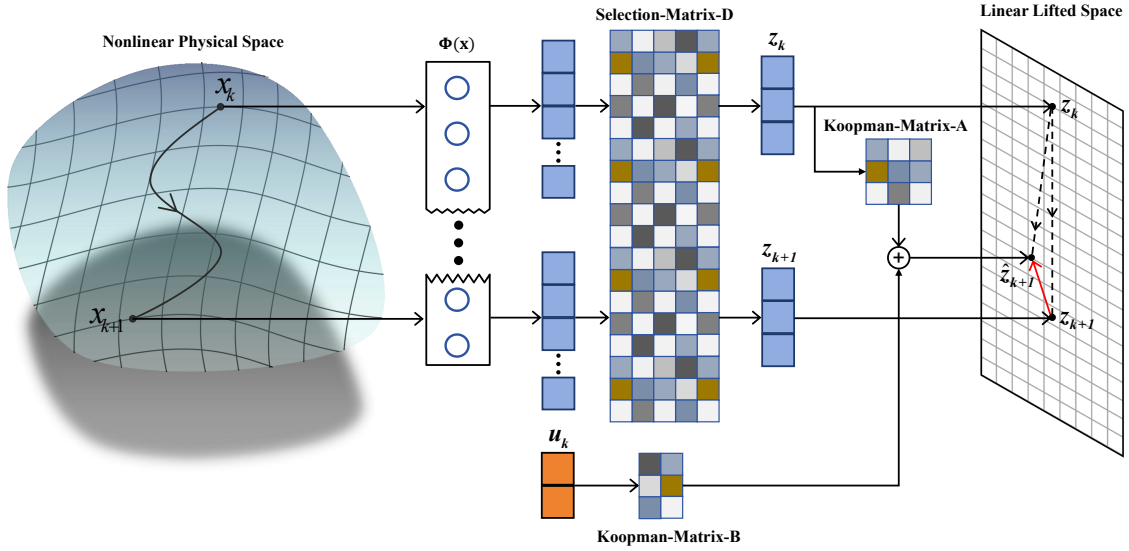


Figure 1: Koopman representation learning with the learnable feature selection mechanism.

Definition 3.1 (Feature-selected augmented state). *Given the candidate library $\Phi(\mathbf{x}_k)$ and the selection matrix $\mathbf{D}$, the compact feature vector and the augmented state are defined as*

$$\boldsymbol{\eta}_k = \mathbf{D}\Phi(\mathbf{x}_k), \quad \mathbf{z}_k = [\mathbf{x}_k^\top, \boldsymbol{\eta}_k^\top]^\top \in \mathbb{R}^{n+N}. \quad (4)$$

Here, $\boldsymbol{\eta}_k$ preserves selected combinations of the overcomplete candidate functions, while the original physical state remains explicitly embedded in $\mathbf{z}_k$.

Based on this augmented representation, we seek the following linear evolution model in the lifted space:

$$\mathbf{z}_{k+1} = \mathbf{A}\mathbf{z}_k + \mathbf{B}\mathbf{u}_k, \quad (5)$$

where $\mathbf{A} \in \mathbb{R}^{(n+N) \times (n+N)}$ is the Koopman matrix governing the linear dynamics in the lifted space, $\mathbf{B} \in \mathbb{R}^{(n+N) \times m}$ is the input matrix relating the control input to the augmented state evolution, and $\mathbf{K} \in \mathbb{R}^{(n+N) \times (n+N+m)}$ is the augmented Koopman matrix defined as $\mathbf{K} \triangleq [\mathbf{A}, \mathbf{B}]$. This structure captures the nonlinear dynamics of the system through $\mathbf{D}\Phi$, while simultaneously allowing lossless recovery of the physical state via the projection matrix $\mathbf{C} = [\mathbf{I}_n, \mathbf{0}_{n \times N}]$ according to

$$\mathbf{x}_k = \mathbf{C}\mathbf{z}_k, \quad (6)$$

which facilitates the subsequent full-state feedback design. Figure 1 illustrates the proposed Koopman representation learning framework.

Given a trajectory dataset $\{\mathbf{x}_k, \mathbf{u}_k\}_{k=1}^{T+1}$, where $\mathbf{x}_k \in \mathbb{R}^n$ represents the robot state (e.g., joint positions and velocities) and $\mathbf{u}_k \in \mathbb{R}^m$ denotes the control torque applied to each joint, which yields T one-step transition pairs, we construct the state snapshot matrices $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T]$ and $\mathbf{Y} = [\mathbf{x}_2, \dots, \mathbf{x}_{T+1}]$, the input matrix $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_T]$, and the feature matrices $\mathbf{X}_\Phi = [\Phi(\mathbf{x}_1), \dots, \Phi(\mathbf{x}_T)]$ and $\mathbf{Y}_\Phi = [\Phi(\mathbf{x}_2), \dots, \Phi(\mathbf{x}_{T+1})]$. For a given $\mathbf{D}$, the lifted state data

matrices are defined as

$$\mathbf{Z}(\mathbf{D}) = \begin{bmatrix} \mathbf{X} \\ \mathbf{D}\mathbf{X}_\Phi \end{bmatrix}, \quad \mathbf{Z}^+(\mathbf{D}) = \begin{bmatrix} \mathbf{Y} \\ \mathbf{D}\mathbf{Y}_\Phi \end{bmatrix}. \quad (7)$$

We formulate the joint optimization problem as

$$\begin{aligned} L(\mathbf{K}, \mathbf{D}) &= \frac{1}{2} \left\| \mathbf{Z}^+(\mathbf{D}) - \mathbf{K} \begin{bmatrix} \mathbf{Z}(\mathbf{D}) \\ \mathbf{U} \end{bmatrix} \right\|_F^2 \\ &\quad + \lambda \|\mathbf{D}\|_1 + \beta \|\mathbf{K}\|_F^2, \\ (\mathbf{K}^*, \mathbf{D}^*) &= \arg \min_{\mathbf{K}, \mathbf{D}} L(\mathbf{K}, \mathbf{D}). \end{aligned} \quad (8)$$

where $\lambda > 0$ and $\beta > 0$ are the regularization coefficients. The ℓ_1 regularization on $\mathbf{D}$ promotes sparse linear combinations of informative candidate functions from the overcomplete candidate function library. Meanwhile, the objective function enforces accurate one-step prediction of both the physical state and the projected feature components in the lifted state, thereby discouraging trivial solutions for $\mathbf{D}$. When $\mathbf{D} = \mathbf{I}_p$ and the regularization terms $\lambda \|\mathbf{D}\|_1$ and $\beta \|\mathbf{K}\|_F^2$ are removed, the formulation reduces to standard EDMD.

Since the optimization problem in (8) is nonconvex in the joint variables $(\mathbf{K}, \mathbf{D})$, we adopt a two-block coordinate descent (BCD) algorithm [22] to solve it by alternately updating the two variable blocks. Let $(\mathbf{K}^{(t)}, \mathbf{D}^{(t)})$ denote the iterates at iteration t , and define the corresponding objective value as

$$L^{(t)} := L(\mathbf{K}^{(t)}, \mathbf{D}^{(t)}).$$

At each iteration, the two variable blocks are updated alternately by solving a strongly convex $\mathbf{K}$ -subproblem and a convex $\mathbf{D}$ -subproblem.

1) **K-subproblem:** With $\mathbf{D}^{(t)}$ fixed, define

$$\mathbf{Y}_{\text{aug}}^{(t)} := \mathbf{Z}^+(\mathbf{D}^{(t)}), \quad \mathbf{W}^{(t)} := \begin{bmatrix} \mathbf{Z}(\mathbf{D}^{(t)}) \\ \mathbf{U} \end{bmatrix}.$$

Then, the $\mathbf{K}$ -subproblem is given by

$$\min_{\mathbf{K}} \frac{1}{2} \|\mathbf{Y}_{\text{aug}}^{(t)} - \mathbf{K}\mathbf{W}^{(t)}\|_F^2 + \beta \|\mathbf{K}\|_F^2.$$

This is a standard ridge regression problem. Setting the gradient with respect to $\mathbf{K}$ to zero gives

$$-(\mathbf{Y}_{\text{aug}}^{(t)} - \mathbf{K}\mathbf{W}^{(t)})\mathbf{W}^{(t)\top} + 2\beta\mathbf{K} = \mathbf{0},$$

from which the closed-form update is derived as

$$\mathbf{K}^{(t+1)} = \mathbf{Z}^+(\mathbf{D}^{(t)})\mathbf{W}^{(t)\top} (\mathbf{W}^{(t)}\mathbf{W}^{(t)\top} + 2\beta\mathbf{I})^{-1}. \quad (9)$$

For the $\mathbf{K}$ update, the ridge regularization makes the subproblem strongly convex for any fixed $\mathbf{D}^{(t)}$ and $\beta > 0$, so the minimizer in (9) is unique.

2) *D-subproblem*: With $\mathbf{K}^{(t+1)}$ fixed, the $\mathbf{D}$ -subproblem is formulated as

$$\mathbf{D}^{(t+1)} = \arg \min_{\mathbf{D}} J^{(t)}(\mathbf{D}), \quad (10)$$

where

$$J^{(t)}(\mathbf{D}) = \frac{1}{2} \left\| \mathbf{Z}^+(\mathbf{D}) - \mathbf{K}^{(t+1)} \begin{bmatrix} \mathbf{Z}(\mathbf{D}) \\ \mathbf{U} \end{bmatrix} \right\|_F^2 + \lambda \|\mathbf{D}\|_1. \quad (11)$$

To see this, we write $\mathbf{K}^{(t+1)} = [\mathbf{A}^{(t+1)}, \mathbf{B}^{(t+1)}]$ and partition $\mathbf{A}^{(t+1)}$ according to the augmented state structure. Since both $\mathbf{Z}(\mathbf{D})$ and $\mathbf{Z}^+(\mathbf{D})$ depend affinely on $\mathbf{D}$, the residual

$$\mathbf{Z}^+(\mathbf{D}) - \mathbf{K}^{(t+1)} \begin{bmatrix} \mathbf{Z}(\mathbf{D}) \\ \mathbf{U} \end{bmatrix}$$

is affine in $\mathbf{D}$. Hence, the data-fitting term in (11) is a convex quadratic function of $\mathbf{D}$, and the subproblem can be efficiently solved using proximal gradient methods [1].

For the $\mathbf{D}$ update, the subproblem in (10) is a convex composite optimization problem consisting of a quadratic data-fitting term and a separable ℓ_1 regularizer. Hence, it can be solved by standard proximal gradient updates.

Let $T_{\max}$ denote the maximum number of iterations and let $\varepsilon > 0$ denote the convergence tolerance. The above block structure leads to the BCD procedure summarized in Algorithm 1. Its convergence property is stated as follows.

Theorem 3.1 (Convergence of the BCD Algorithm). *Suppose that $\lambda > 0$ and $\beta > 0$, and that each subproblem is solved to optimality at every iteration. Then the sequence $\{(\mathbf{K}^{(t)}, \mathbf{D}^{(t)})\}$ generated by Algorithm 1 satisfies:*

1. *The objective values $L^{(t)}$ are monotonically nonincreasing and converge to a finite limit;*
2. *The iterates $\{(\mathbf{K}^{(t)}, \mathbf{D}^{(t)})\}$ remain bounded;*
3. *Every accumulation point of the sequence is a first-order stationary point of (8).*

Proof. First, since each block is minimized exactly while the other block is fixed, the two successive updates in Algorithm 1 yield

$$\begin{aligned} L(\mathbf{K}^{(t+1)}, \mathbf{D}^{(t)}) &\leq L(\mathbf{K}^{(t)}, \mathbf{D}^{(t)}), \\ L(\mathbf{K}^{(t+1)}, \mathbf{D}^{(t+1)}) &\leq L(\mathbf{K}^{(t+1)}, \mathbf{D}^{(t)}). \end{aligned}$$

Therefore, $L^{(t+1)} \leq L^{(t)}$, and the objective sequence is monotonically nonincreasing. Since $L(\mathbf{K}, \mathbf{D}) \geq 0$, it converges to a finite limit.

Second, from $L^{(t)} \leq L^{(0)}$ and the nonnegativity of the data-fitting term, we obtain

$$\beta \|\mathbf{K}^{(t)}\|_F^2 + \lambda \|\mathbf{D}^{(t)}\|_1 \leq L^{(0)}.$$

Hence, both sequences $\{\mathbf{K}^{(t)}\}$ and $\{\mathbf{D}^{(t)}\}$ are bounded, because boundedness in the ℓ_1 norm implies boundedness in the Frobenius norm for finite-dimensional matrices.

Finally, the objective in (8) is the sum of a continuously differentiable data-fitting term and a convex separable nonsmooth regularization term, and each block subproblem is solved to optimality at every iteration. By the standard convergence results for block coordinate descent applied to nonsmooth objectives [22], every accumulation point of $\{(\mathbf{K}^{(t)}, \mathbf{D}^{(t)})\}$ is a first-order stationary point of (8). $\square$

Algorithm 1 Feature-Selection BCD Algorithm

Input: Data $(\mathbf{X}, \mathbf{Y}, \mathbf{U})$, basis library Φ , parameters λ, β , initialization $\mathbf{D}^{(0)}$

Output: Learned $\mathbf{K}^*$ and $\mathbf{D}^*$

- 1: **for** $t = 0, 1, \dots, T_{\max} - 1$ **do**
 - 2: Update $\mathbf{K}^{(t+1)}$ using (9)
 - 3: Update $\mathbf{D}^{(t+1)}$ by solving (10) via a proximal gradient method
 - 4: **if** $|L^{(t+1)} - L^{(t)}| < \varepsilon$ **then**
 - 5: **break**
 - 6: **end if**
 - 7: **end for**
-

3.2. Lumped Disturbance Analysis

The learned Koopman model captures the dominant system dynamics, but the finite-dimensional lifted model inevitably contains approximation residuals. In addition, sensor noise and environmental disturbances may appear during robot operation. We collect these effects into a lumped disturbance term $\mathbf{w}_k$, and the actual evolution in the lifted space is written as

$$\mathbf{z}_{k+1} = \mathbf{A}\mathbf{z}_k + \mathbf{B}\mathbf{u}_k + \mathbf{w}_k, \quad (12)$$

where $\mathbf{A}$ and $\mathbf{B}$ are the learned Koopman state-transition matrix and input matrix, respectively, obtained from Algorithm 1.

For compact state-input sets, the Koopman approximation error is theoretically bounded [8]. Sensor noise and environmental disturbances are bounded in practical robotic systems due to hardware limits and operating conditions. This motivates the following assumption.

Assumption 3.1 (Bounded lumped disturbance). *There exists a constant $\delta_w > 0$ such that the lumped disturbance in (12) satisfies*

$$\|\mathbf{w}_k\| \leq \delta_w, \quad \forall k. \quad (13)$$

Assumption 3.1 links the learned Koopman model to the robust tracking analysis. A smaller approximation residual corresponds to a smaller admissible disturbance level, which is reflected in the ultimate tracking bound derived in Section 4.2.

4. LQR Control Based on Koopman Model

This section designs a feedforward-feedback Koopman-LQR controller for the disturbed lifted model (12) and establishes the ultimate boundedness of the closed-loop tracking error.

4.1. Feedforward-Feedback Robust Controller Design

To achieve robust trajectory tracking, we design the controller based on the disturbed augmented linear model (12) obtained from the learned Koopman model in Section 3. First, let $\{\mathbf{x}_k^{\text{ref}}\}_{k=0}^{\infty}$ denote the desired reference trajectory in the physical state space. By lifting this trajectory through the feature mapping and the learned selection matrix $\mathbf{D}^*$, we obtain the corresponding lifted reference trajectory:

$$\mathbf{z}_k^{\text{ref}} = [\mathbf{x}_k^{\text{ref}^\top}, (\mathbf{D}^* \Phi(\mathbf{x}_k^{\text{ref}}))^\top]^\top. \quad (14)$$

To balance the tracking accuracy and the control efforts, we define the following infinite-horizon quadratic cost function:

$$J = \sum_{k=0}^{\infty} ((\mathbf{z}_k - \mathbf{z}_k^{\text{ref}})^\top \mathbf{Q} (\mathbf{z}_k - \mathbf{z}_k^{\text{ref}}) + \mathbf{u}_k^\top \mathbf{R} \mathbf{u}_k), \quad (15)$$

where $\mathbf{Q} = \mathbf{C}^\top \mathbf{Q}_x \mathbf{C} \geq 0$ is the lifted-state weight matrix, $\mathbf{Q}_x$ is the weight matrix in the original state space, and $\mathbf{R} > 0$ is the control input weight matrix.

Assume that the physical system is controllable and that the training data are sufficiently informative such that the learned linear model $(\mathbf{A}, \mathbf{B})$ is stabilizable. Under these conditions, the standard LQR theory yields a unique optimal state-feedback gain $\mathbf{K}_{\text{lqr}}$, which renders the closed-loop matrix $\mathbf{A}_{\text{cl}} = \mathbf{A} - \mathbf{B}\mathbf{K}_{\text{lqr}}$ Schur stable.

To guarantee the trajectory tracking of the robots, we propose the following composite feedforward-feedback control law:

$$\mathbf{u}_k = \mathbf{u}_k^{\text{ff}} - \mathbf{K}_{\text{lqr}}(\mathbf{z}_k - \mathbf{z}_k^{\text{ref}}). \quad (16)$$

The feedforward term $\mathbf{u}_k^{\text{ff}}$ is used to compensate for the known dynamics along the reference trajectory and is computed as

$$\mathbf{u}_k^{\text{ff}} = \mathbf{B}^\dagger (\mathbf{z}_{k+1}^{\text{ref}} - \mathbf{A} \mathbf{z}_k^{\text{ref}}), \quad (17)$$

where $\mathbf{B}^\dagger$ denotes the Moore-Penrose pseudoinverse of $\mathbf{B}$. Figure 2 illustrates the structure of the proposed Koopman-LQR controller.

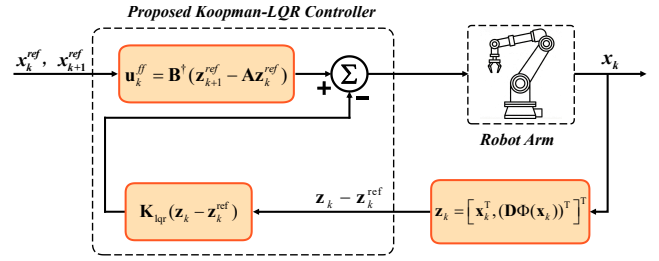


Figure 2: The proposed Koopman-based feedforward-feedback LQR control structure.

4.2. Stability Analysis

Based on the feedforward-feedback controller (16), we present the following stability result.

The key observation is that the feedforward term removes the nominal reference dynamics, whereas the LQR feedback stabilizes the deviation around the lifted reference trajectory. Therefore, the closed-loop tracking error can be analyzed as a stable linear system driven by the lumped disturbance.

Theorem 4.1 (Robust Tracking Performance). *Consider the system (12) under the control law (16). Suppose that Assumption 3.1 holds. Then the closed-loop system achieves ultimately uniformly bounded (UUB) tracking, i.e., there exists $R > 0$ such that, for any initial condition, there exists a finite time K for which $\|\mathbf{e}_k\| \leq R$ for all $k \geq K$.*

Proof. Define the tracking error as $\mathbf{e}_k := \mathbf{z}_k - \mathbf{z}_k^{\text{ref}}$. Substituting the control law (16) into the disturbed augmented model (12), and using the definition of the feedforward term in (17), the closed-loop error dynamics are given by

$$\mathbf{e}_{k+1} = (\mathbf{A} - \mathbf{B}\mathbf{K}_{\text{lqr}})\mathbf{e}_k + \mathbf{w}_k = \mathbf{A}_{\text{cl}}\mathbf{e}_k + \mathbf{w}_k, \quad (18)$$

where $\mathbf{A}_{\text{cl}} := \mathbf{A} - \mathbf{B}\mathbf{K}_{\text{lqr}}$ denotes the closed-loop system matrix. By the stabilizability assumption in Section 4.1 and the standard LQR theory, $\mathbf{A}_{\text{cl}}$ is Schur stable. Hence, there exist positive definite matrices $\mathbf{P} > 0$ and $\mathbf{S} > 0$ satisfying the discrete-time Lyapunov equation

$$\mathbf{A}_{\text{cl}}^\top \mathbf{P} \mathbf{A}_{\text{cl}} - \mathbf{P} = -\mathbf{S}.$$

Consider the Lyapunov function $V_k = \mathbf{e}_k^\top \mathbf{P} \mathbf{e}_k$. Then, along the trajectories of (18),

$$\begin{aligned} \Delta V_k &= V_{k+1} - V_k \\ &= \mathbf{e}_k^\top (\mathbf{A}_{\text{cl}}^\top \mathbf{P} \mathbf{A}_{\text{cl}} - \mathbf{P}) \mathbf{e}_k + 2\mathbf{e}_k^\top \mathbf{A}_{\text{cl}}^\top \mathbf{P} \mathbf{w}_k + \mathbf{w}_k^\top \mathbf{P} \mathbf{w}_k \\ &= -\mathbf{e}_k^\top \mathbf{S} \mathbf{e}_k + 2\mathbf{e}_k^\top \mathbf{A}_{\text{cl}}^\top \mathbf{P} \mathbf{w}_k + \mathbf{w}_k^\top \mathbf{P} \mathbf{w}_k, \end{aligned} \quad (19)$$

where the last equality follows from the discrete-time Lyapunov equation. By the boundedness of $\mathbf{w}_k$ and the Cauchy-Schwarz inequality, it follows that when $\|\mathbf{e}_k\|$ exceeds a threshold R determined by δ_w , one has $\Delta V_k < 0$. Therefore, the error trajectory enters and remains within the bounded set $\mathcal{B}_R = \{\mathbf{e} : \|\mathbf{e}\| \leq R\}$ [2, 6]. $\square$

Since the projection matrix $\mathbf{C}$ satisfies $\|\mathbf{C}\| = 1$, the error bound in the lifted space directly guarantees the tracking accuracy in the original state space:

$$\|\mathbf{x}_k - \mathbf{x}_k^{\text{ref}}\| = \|\mathbf{C}\mathbf{e}_k\| \leq \|\mathbf{e}_k\| \leq R.$$

Furthermore, the steady-state error bound can be upper bounded by a linear function of the disturbance level, i.e.,

$$R \leq c \delta_w,$$

where the constant $c > 0$ depends only on the spectral properties of the closed-loop matrix $\mathbf{A}_{\text{cl}}$ [2]. This relationship connects the Koopman modeling accuracy with the closed-loop performance: reducing the approximation component of $\mathbf{w}_k$ decreases δ_w and hence shrinks the ultimate tracking bound.

5. Experimental Validations

5.1. Experimental Platform and Data Collection

Experiments were conducted on a Geomagic Touch[®] robot arm, whose mechanical structure is shown in Fig. 3. The device has three revolute joints with integrated encoders for real-time measurement of joint position and velocity. The force-feedback stylus at the end-effector was kept fixed during the experiments so that the joint-space dynamics could be isolated.

According to the technical specifications provided by the manufacturer, the motion ranges of joints 1, 2, and 3 are $[-1.3, 0.66]$ rad, $[0.2, 2.0]$ rad, and $[-0.65, 0.87]$ rad, respectively. Joints 2 and 3 are mechanically coupled, so the allowable range of joint 3 is affected when joint 2 approaches its motion limit. To reduce the influence of this coupling, the data collection experiments were conducted to avoid the coupled boundary as much as possible.

During the data collection experiments, continuous random control torques were applied to the three joints to excite the system dynamics. The sampling period was 0.004 s. In total, 20 trajectories were collected, each with a duration of 20 s, containing joint positions, velocities, and the corresponding control inputs. The complete dataset was split into training, validation, and test sets with a ratio of 15:4:1.

The second platform is a cable-driven soft robotic system. Its state is the two-dimensional light-spot position in the xy plane, and its two-dimensional input commands the cable-driven motion of the light spot. The dataset contains 100 trajectories with 300 samples each. A fixed by-trajectory 70/20/10 split is used, and trajectory 66 is used as the displayed modeling test case.

5.2. Model Prediction Accuracy Comparison

For comparison, two Koopman baselines were used: EDMDDL [10], which learns a 60-dimensional dictionary using a five-layer neural network with 512 ReLU neurons per layer, and standard EDMD [24], which directly uses the full 364-dimensional candidate function library. FS-EDMD uses the same candidate function library as standard EDMD but learns a compact selection matrix $\mathbf{D} \in \mathbb{R}^{60 \times 364}$. All



Figure 3: The Geomagic Touch robot arm.

methods were trained on the same training set. For the robotic arm, evaluation uses an 800-step continuous rollout on an independent test trajectory. For the soft platform, the three models are compared on test trajectory 66 over its full rollout.

Fig. 4 shows the prediction results on the robotic arm. FS-EDMD maintains the closest agreement with the measured joint trajectory over the 3.2 s prediction horizon, whereas EDMDDL gradually drifts and standard EDMD accumulates larger errors.

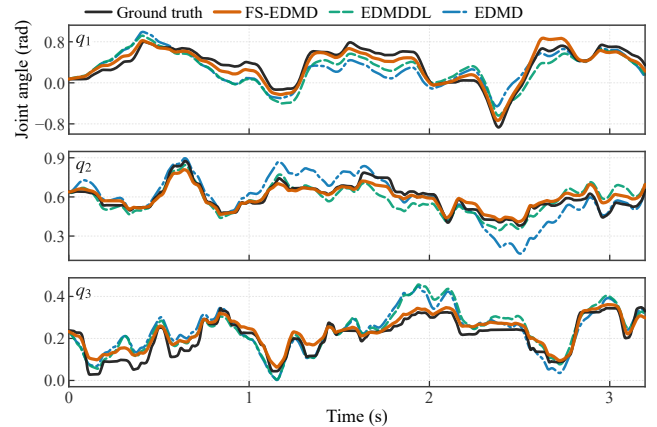


Figure 4: Modeling accuracy on the robotic arm. The panels show Joint 1–3 predictions over the 800-step rollout.

For the soft platform, Fig. 5 shows the x and y predictions on test trajectory 66. The FS-EDMD predictions follow the measured light-spot trajectory more closely than the two baselines over the displayed test trajectory.

Table 1
Modeling Accuracy Comparison

Platform	Method	MSE	RMSE	MAE
Robotic arm	FS-EDMD	0.3520	0.5933	0.3202
	EDMDDL	0.5936	0.7704	0.4550
	EDMD	0.7256	0.8518	0.5022
Soft platform	FS-EDMD	0.6224	0.7889	0.5767
	EDMD	0.8037	0.8965	0.7148
	EDMDDL	1.6711	1.2927	0.9368

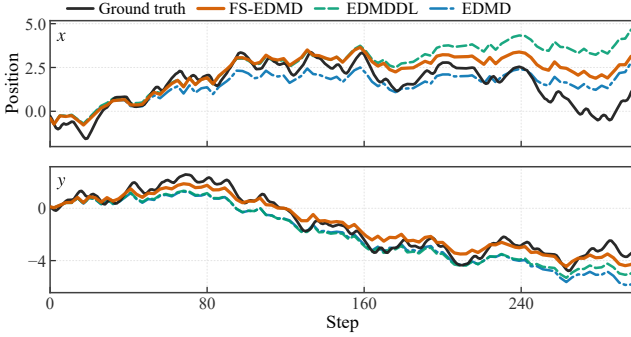


Figure 5: Modeling accuracy on the cable-driven soft platform for test trajectory 66, showing the x and y predictions.

As shown in Table 1, FS-EDMD achieves the lowest full-state prediction errors on the robotic arm, with MSE, RMSE, and MAE of 0.3520, 0.5933, and 0.3202, respectively.¹ These correspond to reductions of 40.7%, 23.0%, and 29.6% over EDMDDL, and 51.5%, 30.3%, and 36.2% over standard EDMD. On the soft platform, FS-EDMD also achieves the lowest errors on test trajectory 66, with an MSE/RMSE/MAE of 0.6224/0.7889/0.5767. These correspond to reductions of 22.6%/12.0%/19.3% over standard EDMD and 62.8%/39.0%/38.4% over EDMDDL.

5.3. Trajectory Tracking Control Performance Evaluation

Trajectory tracking experiments were conducted in the joint space using sinusoidal and irregular references. The Koopman-LQR controller was implemented with the FS-EDMD model and the feedforward-feedback structure in Section 4. The weighting matrices were set to

$$\mathbf{Q} = \text{diag}(100, 10, 100, 10, 100, 10), \quad \mathbf{R} = \mathbf{I}_3,$$

and a PID controller with empirically tuned gains $K_p = 25$, $K_i = 0.1$, and $K_d = 2.5$ was used as the baseline. All tracking metrics were computed over the common 10–40 s evaluation window after controller initialization.

$$^1\text{MSE} = \frac{1}{T} \sum_{k=1}^T \|\mathbf{x}_k - \hat{\mathbf{x}}_k\|^2, \quad \text{RMSE} = \sqrt{\text{MSE}},$$

MAE = $\frac{1}{T} \sum_{k=1}^T \|\mathbf{x}_k - \hat{\mathbf{x}}_k\|$, where $\hat{\mathbf{x}}_k$ denotes the predicted state at time step k . Robotic-arm metrics cover joint positions and velocities, whereas soft-platform metrics cover the xy light-spot position.

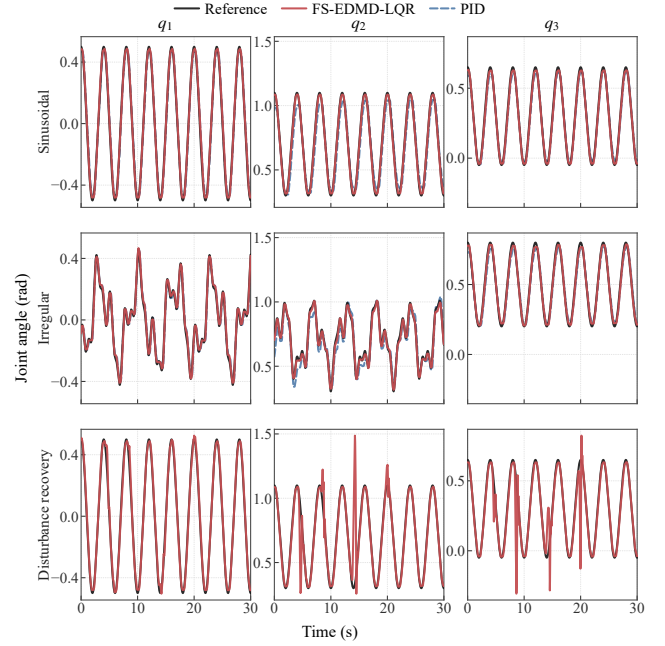


Figure 6: Robotic-arm trajectory tracking under sinusoidal, irregular, and disturbance-recovery references. Rows correspond to the three experiments and columns correspond to Joint 1–3. The first two rows compare Reference, FS-EDMD-LQR, and PID; the third row shows the disturbance-recovery response of FS-EDMD-LQR.

For sinusoidal tracking, the reference trajectories for the three joints were defined as

$$q_i^{\text{ref}}(t) = A_i \sin(\omega_i t) + b_i, \quad i = 1, 2, 3, \quad (20)$$

where q_i^{ref} denotes the desired joint position. The amplitudes were set to $A_1 = 0.5$ rad, $A_2 = 0.4$ rad, and $A_3 = 0.35$ rad, with $f_i = 0.25$ Hz and $b_i = 0$ for all joints. The sinusoidal row of Fig. 6 shows that FS-EDMD-LQR tracks the references more accurately than PID, especially on joints 2 and 3.

For irregular tracking, joints 1 and 2 used composite references formed by the superposition of three sinusoidal components:

$$q_j^{\text{ref}}(t) = \sum_{l=1}^3 A_{j,l} \sin(2\pi f_{j,l} t + \phi_{j,l}), \quad j = 1, 2, \quad (21)$$

where $A_{j,l}$, $f_{j,l}$, and $\phi_{j,l}$ denote the amplitude, frequency, and phase of the l -th component, respectively. The component frequencies were (0.15, 0.40, 0.80) Hz for joint 1 and (0.18, 0.45, 0.90) Hz for joint 2, with amplitudes and phase shifts selected within the experimentally validated operating range. Due to the mechanical coupling between joints 2 and 3, joint 3 used $q_3^{\text{ref}}(t) = 0.35 \sin(2\pi \cdot 0.25t)$.

The irregular row of Fig. 6 shows that FS-EDMD-LQR follows the composite reference more accurately during fast transitions, whereas PID produces larger tracking errors.

Table 2
Trajectory Tracking Performance

Type	Controller	RMSE	MAE
Arm: sinusoidal	FS-EDMD-LQR	0.0154	0.0136
	PID	0.0475	0.0402
Arm: irregular	FS-EDMD-LQR	0.0175	0.0148
	PID	0.0413	0.0337
Arm: disturbance recovery	FS-EDMD-LQR	0.0958	0.0298
	PID	0.6645	0.5468
Soft: triangle	FS-EDMD-LQR	0.1749	0.1703
	PID	0.6645	0.5468
Soft: Lissajous	FS-EDMD-LQR	0.1266	0.1172
	PID	1.7883	1.6710

Robustness was also evaluated within the same arm-tracking setup using the sinusoidal reference and the same controller parameters. During the 30 s experiment, an operator manually disturbed the robotic arm by briefly pushing and pulling the links associated with the three joints, thereby introducing unknown bounded joint torques. In the disturbance-recovery row of Fig. 6, the joint trajectories exhibit temporary deviations from the references during the disturbances but recover smoothly after the disturbances are removed and remain bounded throughout the experiment.

Table 2 summarizes the tracking results.²FS-EDMD-LQR reduces the RMSE²/MAE by 67.5%/66.2% for sinusoidal tracking and by 57.7%/56.0% for irregular tracking, confirming its advantage over PID.

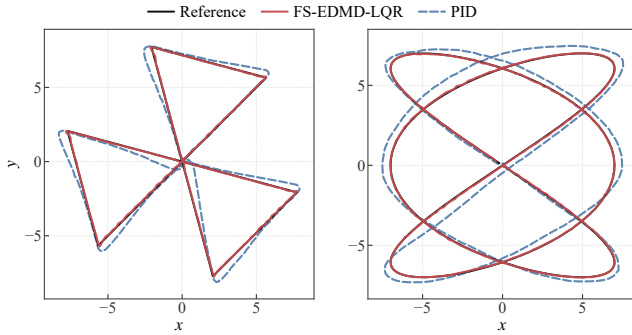


Figure 7: Soft-platform trajectory tracking on the triangle and Lissajous references. The panels compare the complete Reference, FS-EDMD-LQR, and PID trajectories.

For the soft platform, the PID baseline used $K_p = 20.00$, $K_i = 1.00$, and $K_d = 10.00$, and the FS-EDMD-LQR controller used $\mathbf{Q}_x = \text{diag}(250, 250)$ and $\mathbf{R} = \mathbf{I}_2$. Both controllers used the same reference trajectory, initial condition, sampling period, and endpoint condition. To isolate

²For the robotic arm, $\text{RMSE} = \sqrt{\frac{1}{3T} \sum_{i=1}^3 \sum_{k=1}^T (q_i(k) - q_i^{\text{ref}}(k))^2}$ and $\text{MAE} = \frac{1}{3T} \sum_{i=1}^3 \sum_{k=1}^T |q_i(k) - q_i^{\text{ref}}(k)|$. For the soft platform, $\text{RMSE} = \sqrt{\frac{1}{T} \sum_{k=1}^T \|\mathbf{p}_k - \mathbf{p}_k^{\text{ref}}\|_2^2}$ and $\text{MAE} = \frac{1}{T} \sum_{k=1}^T \|\mathbf{p}_k - \mathbf{p}_k^{\text{ref}}\|_2$. Robotic-arm errors are expressed in rad, whereas soft-platform errors use the recorded light-spot position coordinates.

trajectory tracking from the initial movement toward the reference start, RMSE and MAE were computed over the same samples after a common 30-step approach interval. Fig. 7 shows that FS-EDMD-LQR follows both the triangle and Lissajous references more closely than PID. As reported in Table 2, FS-EDMD-LQR reduces the RMSE²/MAE by 73.7%/68.9% on the triangle trajectory and by 92.9%/93.0% on the Lissajous trajectory.

6. Conclusion

This paper addressed the dictionary-design tradeoff in EDMD through a learnable feature-selection matrix. Starting from an overcomplete physics-informed candidate library, the proposed method learns compact feature combinations and updates the lifted dynamics jointly with the selection matrix. A feedforward-feedback Koopman-LQR controller was then designed by treating finite-dimensional approximation errors and external uncertainties as bounded lumped disturbances. Lyapunov analysis establishes ultimate uniform boundedness of the closed-loop tracking error. Experiments on a three-degree-of-freedom rigid robot arm and a cable-driven soft platform show lower multi-step prediction errors than EDMD and EDMDDL and lower tracking errors than PID over the tested references. A separate disturbance-recovery test on the robotic arm further shows bounded deviations and smooth recovery. These results are limited to the tested operating ranges and bounded-disturbance setting. Future work will consider online adaptation and model predictive control to account explicitly for state and input constraints.

CRediT authorship contribution statement

First Author: Conceptualization, Methodology, Software, Writing – original draft.

References

- [1] Beck, A., Teboulle, M., 2009. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM Journal on Imaging Sciences* 2, 183–202. doi:10.1137/080716542.
- [2] Blanchini, F., Miani, S., 2008. *Set-Theoretic Methods in Control*. Birkhäuser, Boston, MA, USA. doi:10.1007/978-0-8176-4606-6.
- [3] Brunke, L., Greeff, M., Hall, A.W., Yuan, Z., Zhou, S., Panerati, J., Schoellig, A.P., 2022. Safe learning in robotics: From learning-based control to safe reinforcement learning. *Annual Review of Control, Robotics, and Autonomous Systems* 5, 411–444. doi:10.1146/annurev-control-042920-020211.
- [4] Brunton, S.L., Brunton, B.W., Proctor, J.L., Kutz, J.N., 2016. Koopman invariant subspaces and finite linear representations of nonlinear dynamical systems for control. *PLOS ONE* 11, e0150171. doi:10.1371/journal.pone.0150171.
- [5] Brunton, S.L., Budišić, M., Kaiser, E., Kutz, J.N., 2022. Modern Koopman theory for dynamical systems. *SIAM Review* 64, 229–340. doi:10.1137/21M1401243.
- [6] Khalil, H.K., 2002. *Nonlinear Systems*. 3rd ed., Prentice Hall, Upper Saddle River, NJ, USA.
- [7] Koopman, B.O., 1931. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences of the United States of America* 17, 315–318. doi:10.1073/pnas.17.5.315.

- [8] Korda, M., Mezić, I., 2018a. Linear predictors for nonlinear dynamical systems: Koopman operator meets model predictive control. *Automatica* 93, 149–160. doi:10.1016/j.automatica.2018.03.046.
- [9] Korda, M., Mezić, I., 2018b. On convergence of extended dynamic mode decomposition to the Koopman operator. *Journal of Nonlinear Science* 28, 687–710. doi:10.1007/s00332-017-9423-0.
- [10] Li, Q., Dietrich, F., Bollt, E.M., Kevrekidis, I.G., 2017. Extended dynamic mode decomposition with dictionary learning: A data-driven adaptive spectral decomposition of the Koopman operator. *Chaos: An Interdisciplinary Journal of Nonlinear Science* 27, 103111. doi:10.1063/1.4993854.
- [11] Li, Y., Li, C., Liu, K., Dong, J., Johansson, R., 2026. Event-triggered control and communication for single-master multislave teleoperation systems with try-once-discard protocol. *IEEE Transactions on Cybernetics* 56, 1117–1130. doi:10.1109/TCYB.2025.3621182.
- [12] Li, Y., Su, R., 2026. Safety-guaranteed adaptive control of teleoperation systems under event-triggered mechanisms. *IEEE Transactions on Control Systems Technology* 34, 1649–1656. doi:10.1109/TCST.2026.3662978.
- [13] Lusch, B., Kutz, J.N., Brunton, S.L., 2018. Deep learning for universal linear embeddings of nonlinear dynamics. *Nature Communications* 9, 4950. doi:10.1038/s41467-018-07210-0.
- [14] Mamakoukas, G., Castaño, M.L., Tan, X., Murphey, T.D., 2021. Derivative-based Koopman operators for real-time control of robotic systems. *IEEE Transactions on Robotics* 37, 2173–2192. doi:10.1109/TRO.2021.3076581.
- [15] Mauroy, A., Goncalves, J., 2020. Koopman-based lifting techniques for nonlinear systems identification. *IEEE Transactions on Automatic Control* 65, 2550–2565. doi:10.1109/TAC.2019.2941433.
- [16] Proctor, J.L., Brunton, S.L., Kutz, J.N., 2016. Dynamic mode decomposition with control. *SIAM Journal on Applied Dynamical Systems* 15, 142–161. doi:10.1137/15M1013857.
- [17] Ren, C., Jiang, H., Li, C., Sun, W., Ma, S., 2023. Koopman-operator-based robust data-driven control for wheeled mobile robots. *IEEE/ASME Transactions on Mechatronics* 28, 461–472. doi:10.1109/TMECH.2022.3203518.
- [18] Santina, C.D., Duriez, C., Rus, D., 2023. Model-based control of soft robots: A survey of the state of the art and open challenges. *IEEE Control Systems Magazine* 43, 30–65. doi:10.1109/MCS.2023.3253419.
- [19] Shi, H., Meng, M.Q.H., 2022. Deep Koopman operator with control for nonlinear systems. *IEEE Robotics and Automation Letters* 7, 7700–7707. doi:10.1109/LRA.2022.3184036.
- [20] Shi, L., Karydis, K., 2022. ACD-EDMD: Analytical construction for dictionaries of lifting functions in Koopman operator-based nonlinear robotic systems. *IEEE Robotics and Automation Letters* 7, 906–913. doi:10.1109/LRA.2021.3133001.
- [21] Tang, C., Abbatematteo, B., Hu, J., Chandra, R., Martín-Martín, R., Stone, P., 2025. Deep reinforcement learning for robotics: A survey of real-world successes. *Annual Review of Control, Robotics, and Autonomous Systems* 8, 153–188. doi:10.1146/annurev-control-030323-022510.
- [22] Tseng, P., 2001. Convergence of a block coordinate descent method for nondifferentiable minimization. *Journal of Optimization Theory and Applications* 109, 475–494. doi:10.1023/A:1017501703105.
- [23] Wang, J., Xu, B., Lai, J., Wang, Y., Hu, C., Li, H., Song, A., 2023. An improved Koopman-MPC framework for data-driven modeling and control of soft actuators. *IEEE Robotics and Automation Letters* 8, 616–623. doi:10.1109/LRA.2022.3229235.
- [24] Williams, M.O., Kevrekidis, I.G., Rowley, C.W., 2015. A data-driven approximation of the Koopman operator: Extending dynamic mode decomposition. *Journal of Nonlinear Science* 25, 1307–1346. doi:10.1007/s00332-015-9258-5.
- [25] Zhang, X., Pan, W., Scattolini, R., Yu, S., Xu, X., 2022. Robust tube-based model predictive control with Koopman operators. *Automatica* 137, 110114. doi:10.1016/j.automatica.2021.110114.